

Scar tissue can narrow the outlet of the bladder after prostate treatment. It slows or blocks the urine stream. Treatment can open it. Because the scar sits next to the muscle that keeps you dry, protecting bladder control is part of every treatment choice.

About This Condition

Where the scar forms depends on the treatment you had:

- **Bladder neck contracture:** after surgery for an enlarged prostate, such as TURP, HoLEP or laser surgery.
- **Anastomotic stenosis:** after prostate removal for cancer, where the bladder was joined to the urethra. It usually appears in the first year.
- **Stenosis after radiation:** often in the deep urethra near the control muscle. It can appear years after treatment.

Signs include a weak or slow stream, straining, a feeling of not emptying, urine infections, or being unable to urinate. Tests include a urine test, a flow test, a **camera look** (cystoscopy) and an **X-ray with dye** (urethrogram).

Benefits and Risks in Numbers

about
4 in 10

needed treatment again within 5 years after scope treatment of a narrowed join

about
8 in 10

stayed open after reconstruction surgery. Results are better at the bladder neck

about
1 in 5

had new leaking after surgery to repair a narrowing caused by radiation

Numbers are averages from published studies. Your own risk may differ; ask your team.

Leaking and a Later Sphincter

- A tight scar can act like a cork. Once it is opened, leaking can show up or get worse. Leaking is more likely after radiation.
- The usual order is to **open the outlet first**, check that it stays open, and then treat leaking.
- Leaking that lasts is often treated with an **artificial urinary sphincter**, which has its own handout. Radiation and some repairs raise its risk of problems.

Learn the Terms

Bladder neck

The outlet where the bladder drains into the urethra.

Contracture, stenosis

A narrowing caused by scar tissue.

Posterior urethra

The deep part of the urethra, next to the bladder and the control muscle.

Anastomosis

The join made between the bladder and urethra when the prostate is removed.

Sphincter

The muscle that holds urine in until you choose to go.

Dilation

Stretching the scar open, often with a balloon.

Incision

Cutting the scar open through a scope, with no cut on the skin.

Reconstruction

Surgery that removes or bypasses the scar and rebuilds the outlet.

Artificial urinary sphincter

An implant with a cuff and pump that you control.

Suprapubic catheter

A tube that drains the bladder through the lower belly.

Will it come back? It can, especially after radiation or several scope treatments. If the narrowing returns after one or two scope treatments, or the channel is fully closed, ask about reconstruction rather than repeating the same treatment.

Tests

Flow test, camera look and X-ray show where the scar is and how long.

First treatment

Most narrowings are stretched or cut open through a scope.

If it comes back

Repeat once, or plan reconstruction at a specialist center.

Once it stays open

Leaking, if any, is checked and treated.

Treatment Choices**Scope treatment** FIRST STEP FOR MOST

Stretching, cutting or removing the scar through the urethra. It is the least invasive choice but often needs repeating.

Self-catheterization TO KEEP IT OPEN

Passing a thin tube on a schedule after treatment, or long term if you do not want more surgery. It can become a burden.

Reconstruction FOR SCAR THAT KEEPS COMING BACK

Robotic or open surgery to remove the scar and rebuild the outlet, sometimes with a bladder flap or a cheek graft.

A drainage tube or urinary diversion WHEN DAMAGE IS SEVERE

A suprapubic catheter, or rerouting urine away from the bladder, after severe radiation damage or many failed repairs.

Call the office that treats you if you have:

- A temperature of **100.4 °F (38 °C)** or higher, or chills
- A stream that is slowing again, or new or worse leaking
- Air or stool in the urine, urine from the rectum, or new pain in the pubic bone or groin

Go to an emergency department if:

- You cannot urinate for several hours and your bladder feels full or painful
- Your urine looks like red wine or has clots, or a catheter stops draining
- You cannot reach the office and are worried

Three Things to Remember

1. Scar after prostate treatment can narrow the bladder outlet. It can be opened.
2. Scope treatment comes first for most, but the scar often returns. Reconstruction is the next step.
3. Opening the outlet can bring out leaking. It is treated once the outlet stays open.

How to Get Ready

- A **urine test** checks for infection. Any infection is treated first.
- Tell your team about past radiation and treatments, blood thinners, and how many pads you use now.
- Bring your latest cancer results, such as your PSA.